

TASK 1: Behavioral Scoring Framework

Objective

To design a dynamic engagement score (0–100) that captures a student's weekly learning behaviour and trajectory, rather than relying solely on static academic outcomes.

Data Processing

- Combined multiple VLE clickstream files (10M+ interactions) into a unified dataset
- Merged with activity metadata (activity type)
- Converted timestamps into weekly intervals for temporal analysis

Feature Engineering

Six behavioral features were constructed:

- **Activity Frequency (Clicks per Week)**
Total number of interactions per student per week
Rationale: Higher interaction indicates active participation in learning materials
- **Activity Diversity**
Number of unique activity types accessed per week
Rationale: Students engaging with multiple resource types demonstrate deeper learning involvement
- **Recency Score**
Measures how recently a student was active
Computed using exponential decay based on inactivity days
Rationale: Recent inactivity is a strong early signal of disengagement
- **Submission Behaviour**
Weekly count of assessment submissions
Rationale: Regular submissions indicate discipline and academic engagement
- **Assessment Consistency**
Derived from the variance in student assessment scores
Lower variance = higher consistency
Rationale: Stable performance reflects sustained effort and understanding
- **Weekly Trend Slope**
Rolling slope of activity over a 3-week window
Rationale: Captures whether engagement is increasing or declining over time

Feature Normalization

All features were normalized using MinMax scaling to ensure comparability across different ranges.

Engagement Score Formulation

The final engagement score is a weighted combination of normalized features:

- Frequency (0.25) - primary indicator of activity
- Recency (0.20) - captures current engagement
- Trend (0.15) - captures trajectory
- Diversity (0.15) - depth of interaction
- Submissions (0.15) - academic discipline
- Consistency (0.10) - performance stability

Final score scaled to 0–100

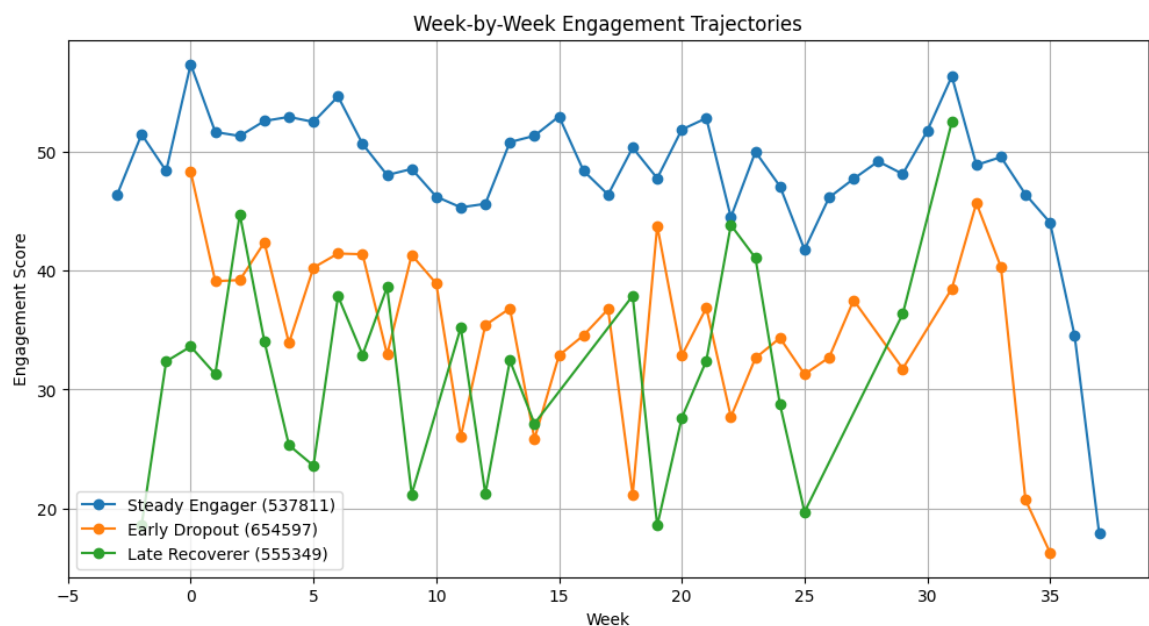
The score is:

Dynamic (week-by-week)

Behaviour-driven (not grade-dependent)

Sensitive to trends (early warning capable)

A combined trajectory plot was generated showing week-by-week engagement scores for all three archetypes.



To validate the relevance of the engineered features, their **average values were analyzed across different final student outcomes** (Distinction, Pass, Fail, Withdrawn).

This analysis ensures that each feature contributes meaningfully to distinguishing between successful and at-risk students.

TASK 2: Predictive Disengagement Model

Objective

To predict whether a student will withdraw or fail (1) versus pass or achieve distinction (0) using only behavioral data available up to Week 6, enabling early intervention.

Data & Feature Design

Behavioral features were aggregated per student using data up to Week 6 :

- Activity intensity: average & total clicks
- Engagement depth: activity diversity
- Temporal behaviour: recency score, trend slope
- Academic signals: submission count, assignment consistency

This ensures no data leakage and reflects real-world deployment constraints.

Target Definition

- 1 - At Risk (Withdraw / Fail)
- 0 - Not At Risk (Pass / Distinction)

Handling Class Imbalance

- Original imbalance (0:11482,1:8456)
- Applied **SMOTE** to balance classes

Improved model sensitivity toward at-risk students.

To capture the complex and multi-dimensional nature of student disengagement, a **custom stacking ensemble model** was designed rather than relying on a single algorithm.

Base Learners (Diverse Learning Perspectives):

- **Random Forest**: Useful for modeling general behavioral patterns
- **XGBoost**: Helps identify edge cases (students close to disengagement)
- **LightGBM**: Handles high-dimensional behavioral signals effectively

Meta Learner: Extra Trees Classifier

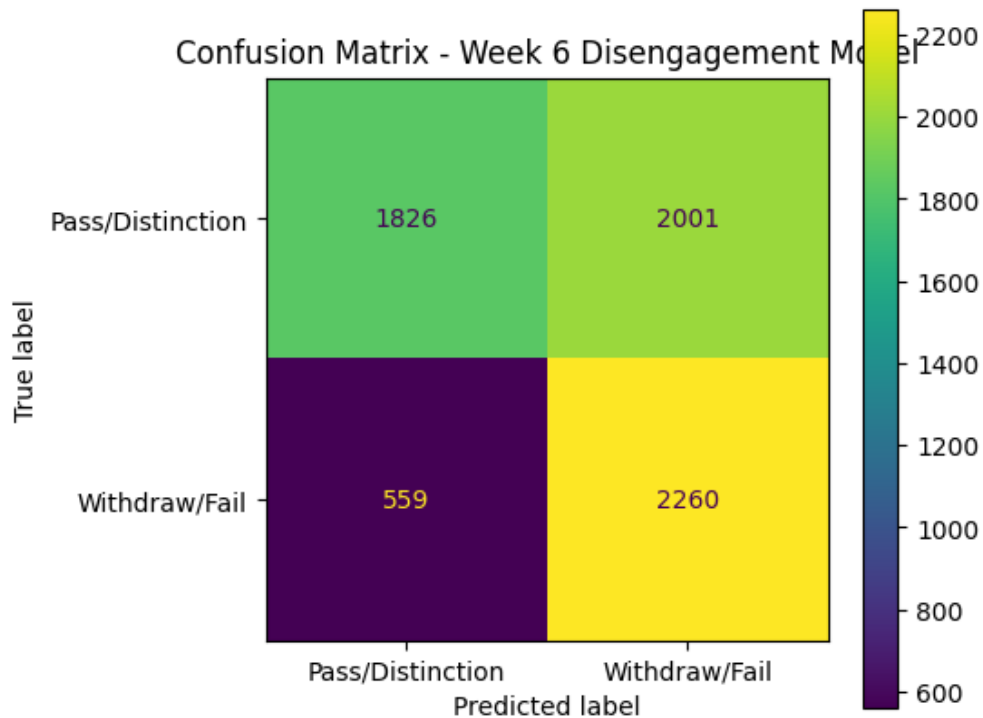
- Receives predictions from all base models
- Learns how to combine their strengths optimally

Threshold Optimization

- Instead of using the default threshold (0.5):
- Threshold was tuned across values
- Final threshold selected: 0.41

This ensures:

- High recall (80%) - most at-risk students are captured
- Acceptable precision - manageable false positives



- High true positive rate (at-risk students detected)
- Some false positives (acceptable trade-off)

Model favours early detection over strict precision

Why Recall is Prioritized

- Missing an at-risk student = **no intervention - dropout/failure**
- False positives are acceptable (advisor review filters them)

Model is intentionally designed to **capture maximum at-risk students early.**

Calibration Analysis

- Predicted probabilities were evaluated using risk buckets.
- High-risk predictions ($\geq 60\%$) closely match actual disengagement
- Model is well-calibrated where decisions matter most

Key Drivers of Disengagement

Permutation importance identified:

- **Assessment Consistency**
 - Irregular performance - higher disengagement risk
- **Total Submissions (Week ≤ 6)**

- Fewer submissions - reduced academic participation
- **Recency Score**
 - Lower recent activity - early disengagement signal

Disengagement is driven by a combination of **behavioral inactivity and academic instability**.

TASK 3: Course Recommendation Engine

Objective

To recommend the top 3 most suitable courses for a student based on their behavioral profile and past learning patterns, supporting data-driven academic guidance.

System Design Overview

Two complementary recommendation approaches were implemented:

- Content-Based Filtering (Student–Course Matching)
- Collaborative Filtering (Peer-Based Recommendations)

This dual approach ensures both personalization and collective intelligence.

Approach 1: Content-Based Recommendation

Method

- Built **student profiles** using: Engagement score, Click activity, Diversity, Recency & Success rate
- Built **course profiles** using aggregated student behavior and outcomes
- Used **cosine similarity** to match student profiles with course profiles

Recommend courses that are behaviorally aligned with the student's strengths

Approach 2: Collaborative Filtering

Method

- Created a **student-course interaction matrix** using a custom rating:

$$Rating = 0.7 \times Engagement\ Score + 30 \times Success$$
- Computed **student-to-student similarity** using cosine similarity
- Recommended courses taken by **similar students**

Cold start strategy

For new students with no history:

A hybrid scoring approach was used:

- 50% → Course success rate
- 30% → Course popularity
- 20% → Average engagement

Evaluation

- A proxy evaluation was performed using **HitRate@3**:
- **HitRate@3 = 0.58**
- Interpretation:
In 58% of cases, the model successfully recommends a course that the student actually succeeded in.

This project develops a three-part system designed to support both students and university staff:

1. Engagement Tracking System

A weekly engagement score (0–100) was created to measure how actively students are interacting with learning materials. This score captures patterns such as:

- Frequency of activity
- Consistency of participation
- Recent engagement behavior

Unlike grades, this score reflects real-time learning behavior, allowing early detection of declining engagement.

2. Early Risk Prediction Model

A predictive model was developed to identify students who are likely to withdraw or fail before Week 6 of the course.

Key outcomes:

- Successfully identifies **over 80% of at-risk students**
- Prioritizes early detection so that **no struggling student is missed**
- The system converts predictions into **clear alerts for academic advisors**, including:
 - Risk level
 - Key reasons for concern
 - Suggested actions (e.g., early outreach, mentoring support)

3. Course Recommendation System

A recommendation engine suggests the **top 3 most suitable courses** for each student based on:

- Their learning behavior
- Performance patterns
- Similar students' outcomes

It also supports new students with no prior data by recommending **high-success, widely taken courses**.

